LAB QUESTIONS – MySQL 21st November 2025

PART 1 — SQL Queries

Q1. Create a table named students with fields:

• stdid INT PRIMARY KEY

• stdname VARCHAR(50)

• age INT

• city VARCHAR(50)

```
Database changed
mysql> CREATE TABLE students (stdid INT PRIMARY KEY, stdname VARCHAR(50), age INT, city VARCHAR(50));
Query OK, 0 rows affected (0.09 sec)
```

Q2. Insert the following records into the students table:

stdid	stdname	age	city
1	Rohan	20	Pune
2	Meera	22	Mumbai
3	Arjun	21	Delhi
4	Kavya	23	Pune
5	Neha	22	Kolkata

```
mysql> INSERT INTO students VALUES (1,'Rohan',20,'Pune'),(2,'Meera',22,'Mumbai'),(3,'Arjun',
21,'Delhi'),(4,'Kavya',23,'Pune'),(5,'Neha',22,'Kolkata');
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

Q3. Display all student records.

```
mysql> select * from students;
+-------+---------+------+---------+
| stdid | stdname | age  | city    |
+-------+---------+------+---------+
|     1 | Rohan   |   20 | Pune    |
|     2 | Meera   |   22 | Mumbai  |
|     3 | Arjun   |   21 | Delhi   |
|     4 | Kavya   |   23 | Pune    |
|     5 | Neha    |   22 | Kolkata |
+-------+---------+------+---------+
5 rows in set (0.01 sec)
```

Q4. Display only the name and age of all students.

```
mysql> select stdname, age from students;
+---------+------+
| stdname | age  |
+---------+------+
| Rohan   |   20 |
| Meera   |   22 |
| Arjun   |   21 |
| Kavya   |   23 |
| Neha    |   22 |
+---------+------+
5 rows in set (0.01 sec)
```

Q4. Display only the name and age of all students.

```
mysql> select stdname, age from students;
+---------+------+
| stdname | age  |
+---------+------+
| Rohan   |   20 |
| Meera   |   22 |
| Arjun   |   21 |
| Kavya   |   23 |
| Neha    |   22 |
+---------+------+
5 rows in set (0.01 sec)
```

Q5. Display students who are from Pune.

```
mysql> select * from students where city='Pune';
+-------+---------+------+------+
| stdid | stdname | age  | city |
+-------+---------+------+------+
|     1 | Rohan   |   20 | Pune |
|     4 | Kavya   |   23 | Pune |
+-------+---------+------+------+
2 rows in set (0.01 sec)
```

Q6. Display students whose age is greater than 21.

```
mysql> select * from students where age > 21;
+-------+---------+------+---------+
| stdid | stdname | age  | city    |
+-------+---------+------+---------+
|     2 | Meera   |   22 | Mumbai  |
|     4 | Kavya   |   23 | Pune    |
|     5 | Neha    |   22 | Kolkata |
+-------+---------+------+---------+
3 rows in set (0.00 sec)
```

Q7. Display students in descending order of age.

```
mysql> select * from students order by age desc;
+-------+---------+------+---------+
| stdid | stdname | age  | city    |
+-------+---------+------+---------+
|     4 | Kavya   |   23 | Pune    |
|     2 | Meera   |   22 | Mumbai  |
|     5 | Neha    |   22 | Kolkata |
|     3 | Arjun   |   21 | Delhi   |
|     1 | Rohan   |   20 | Pune    |
+-------+---------+------+---------+
5 rows in set (0.00 sec)
```

Q 8. Count how many students belong to each city. (Use GROUP BY).

```
mysql> select city, count(*) from students group by city;
+---------+----------+
| city    | count(*) |
+---------+----------+
| Pune    |        2 |
| Mumbai  |        1 |
| Delhi   |        1 |
| Kolkata |        1 |
+---------+----------+
4 rows in set (0.01 sec)
```

Q9. Display students whose name starts with ‘K’. (Use LIKE)

```
mysql> select * from students where stdname like 'k%';
+-------+---------+------+------+
| stdid | stdname | age  | city |
+-------+---------+------+------+
|     4 | Kavya   |   23 | Pune |
+-------+---------+------+------+
1 row in set (0.01 sec)
```

Q10. Delete student whose stdid = 5.

```
mysql> select * from students where stdid = 5;
+-------+---------+------+---------+
| stdid | stdname | age  | city    |
+-------+---------+------+---------+
|     5 | Neha    |   22 | Kolkata |
+-------+---------+------+---------+
1 row in set (0.00 sec)
```

PART 2 — ALTER COMMAND QUESTIONS

Q11. Add a new column contact VARCHAR(15) to the students table.

```
mysql> alter table students add contact varchar(15);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM students;
+-------+---------+------+---------+---------+
| stdid | stdname | age  | city    | contact |
+-------+---------+------+---------+---------+
|     1 | Rohan   |   20 | Pune    | NULL    |
|     2 | Meera   |   22 | Mumbai  | NULL    |
|     3 | Arjun   |   21 | Delhi   | NULL    |
|     4 | Kavya   |   23 | Pune    | NULL    |
|     5 | Neha    |   22 | Kolkata | NULL    |
+-------+---------+------+---------+---------+
5 rows in set (0.00 sec)
```

Q12. Modify the data type of city column to VARCHAR(100).

```
mysql> alter table students modify city varchar(100);
Query OK, 5 rows affected (0.09 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from students;
+-------+---------+------+---------+---------+
| stdid | stdname | age  | city    | contact |
+-------+---------+------+---------+---------+
|     1 | Rohan   |   20 | Pune    | NULL    |
|     2 | Meera   |   22 | Mumbai  | NULL    |
|     3 | Arjun   |   21 | Delhi   | NULL    |
|     4 | Kavya   |   23 | Pune    | NULL    |
|     5 | Neha    |   22 | Kolkata | NULL    |
+-------+---------+------+---------+---------+
5 rows in set (0.00 sec)
```

al

Q13. Rename the column stdname to student_name.

```
mysql> alter table students drop column contact;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from students;
+-------+--------------+------+---------+
| stdid | student_name | age  | city    |
+-------+--------------+------+---------+
|     1 | Rohan        |   20 | Pune    |
|     2 | Meera        |   22 | Mumbai  |
|     3 | Arjun        |   21 | Delhi   |
|     4 | Kavya        |   23 | Pune    |
|     5 | Neha         |   22 | Kolkata |
+-------+--------------+------+---------+
5 rows in set (0.00 sec)
```

Q14. Drop the column contact from the table.

```
mysql> alter table students drop column contact;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from students;
+-------+--------------+------+---------+
| stdid | student_name | age  | city    |
+-------+--------------+------+---------+
|     1 | Rohan        |   20 | Pune    |
|     2 | Meera        |   22 | Mumbai  |
|     3 | Arjun        |   21 | Delhi   |
|     4 | Kavya        |   23 | Pune    |
|     5 | Neha         |   22 | Kolkata |
+-------+--------------+------+---------+
5 rows in set (0.00 sec)
```

Q15. Add a new column gender ENUM('M','F').

```
mysql> alter table students add gender ENUM('M','F');
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from students;
+-------+--------------+------+---------+--------+
| stdid | student_name | age  | city    | gender |
+-------+--------------+------+---------+--------+
|     1 | Rohan        |   20 | Pune    | NULL   |
|     2 | Meera        |   22 | Mumbai  | NULL   |
|     3 | Arjun        |   21 | Delhi   | NULL   |
|     4 | Kavya        |   23 | Pune    | NULL   |
|     5 | Neha         |   22 | Kolkata | NULL   |
+-------+--------------+------+---------+--------+
5 rows in set (0.00 sec)
```

PART 3 — JOIN PRACTICE

Tables:

Table: students

stdid	**Student_name**	**city**
1	**Rohit**	**Pune**
2	**Meera**	**Mumbai**
3	**Arjun**	**Delhi**
4	**Kavya**	**Pune**

Table: marks

stdid	**subject**	**marks**
1	Maths	88
2	Maths	76
3	Maths	92
5	Maths	76

INNER JOIN

Q16. Display student name and marks of only those students who have matching IDs in both tables.

(Students without marks should not appear.)

```
mysql> select student_name, marks from students inner join marks on students
.stdid = marks.stdid;
+--------------+-------+
| student_name | marks |
+--------------+-------+
| Rohan        |    88 |
| Meera        |    76 |
| Arjun        |    92 |
+--------------+-------+
3 rows in set (0.00 sec)
```

LEFT JOIN

Q17. Display all students and their marks.

(If marks not available, show NULL.)

```
mysql> select student_name, marks from students left join marks on students.
stdid = marks.stdid;
+--------------+-------+
| student_name | marks |
+--------------+-------+
| Rohan        |    88 |
| Meera        |    76 |
| Arjun        |    92 |
| Kavya        |  NULL |
+--------------+-------+
4 rows in set (0.01 sec)
```

RIGHT JOIN

Q18. Display all marks records along with student names.

(If student doesn't exist in students table, show NULL.)

```
mysql> select student_name, marks from students right join marks on students
.stdid = marks.stdid;
+--------------+-------+
| student_name | marks |
+--------------+-------+
| Rohan        |    88 |
| Meera        |    76 |
| Arjun        |    92 |
| NULL         |    67 |
+--------------+-------+
4 rows in set (0.00 sec)
```

CROSS JOIN

Q19. Display all possible combinations of students and subjects.

(Use CROSS JOIN between students and marks table to show every pair.)

```
mysql> select students.stdid, student_name, marks.subject, marks.marks from
students cross join marks;
+-------+--------------+---------+-------+
| stdid | student_name | subject | marks |
+-------+--------------+---------+-------+
|     4 | Kavya        | Maths   |    88 |
|     3 | Arjun        | Maths   |    88 |
|     2 | Meera        | Maths   |    88 |
|     1 | Rohan        | Maths   |    88 |
|     4 | Kavya        | Maths   |    76 |
|     3 | Arjun        | Maths   |    76 |
|     2 | Meera        | Maths   |    76 |
|     1 | Rohan        | Maths   |    76 |
|     4 | Kavya        | Maths   |    92 |
|     3 | Arjun        | Maths   |    92 |
|     2 | Meera        | Maths   |    92 |
|     1 | Rohan        | Maths   |    92 |
|     4 | Kavya        | Maths   |    67 |
|     3 | Arjun        | Maths   |    67 |
|     2 | Meera        | Maths   |    67 |
|     1 | Rohan        | Maths   |    67 |
+-------+--------------+---------+-------+
16 rows in set (0.00 sec)
```

JOIN with Filtering

Q20. Using INNER JOIN, display students who scored more than 80.

```
mysql> select student_name, marks from students inner join marks on students
.stdid = marks.stdid where marks > 80;
+--------------+-------+
| student_name | marks |
+--------------+-------+
| Rohan        |    88 |
| Arjun        |    92 |
+--------------+-------+
2 rows in set (0.00 sec)
```